

Time to Stable Operating Temperature

PrimeX 13, PrimeX 22 and PrimeX 41

OptiTrack QA and R&D

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Summary

Every OptiTrack camera warms up under use, eventually reaching a steady temperature. Because the camera body and lens move slightly as they warm, calibration and activities that require accuracy and precision should be performed after the camera has reached its stable temperature. This report provides a general idea of how long each camera model needs to run to reach its stable operating temperature within a temperature-controlled environment at default settings (120hz, LED on, 250 microsecond exposure).

The table below summarizes the results.

Camera	Start condition	Time to stable	Stable temperature	Total rise
PrimeX 13	Hibernation Start	50 min	51.6 °C	14.8 °C
PrimeX 13	Cold Start	49 min	50.9 °C	23.6 °C
PrimeX 22	Hibernation Start	39 min	45.9 °C	6.4 °C
PrimeX 22	Cold Start	80 min	46.4 °C	19.1 °C
PrimeX 41	Hibernation Start	61 min	45.3 °C	7.8 °C
PrimeX 41	Cold Start	59 min	44.6 °C	14.4 °C

What was measured

Cameras of each model were run as a set, with all cameras configured identically and started simultaneously. Temperature readings were recorded continuously throughout the run via the Camera SDK with a resolution of 1°C.

Run	Start	Cameras	Frame rate	Ambient	IR LEDs	Duration
PrimeX 13	Hibernation Start	16	120 Hz	20 °C	On	83 min
PrimeX 13	Cold Start	16	120 Hz	20 °C	On	107 min
PrimeX 22	Hibernation Start	16	120 Hz	20 °C	On	90 min
PrimeX 22	Cold Start	16	120 Hz	20 °C	On	120 min
PrimeX 41	Hibernation Start	16	120 Hz	20 °C	On	90 min
PrimeX 41	Cold Start	16	120 Hz	20 °C	On	113 min

Results

Hibernation Start

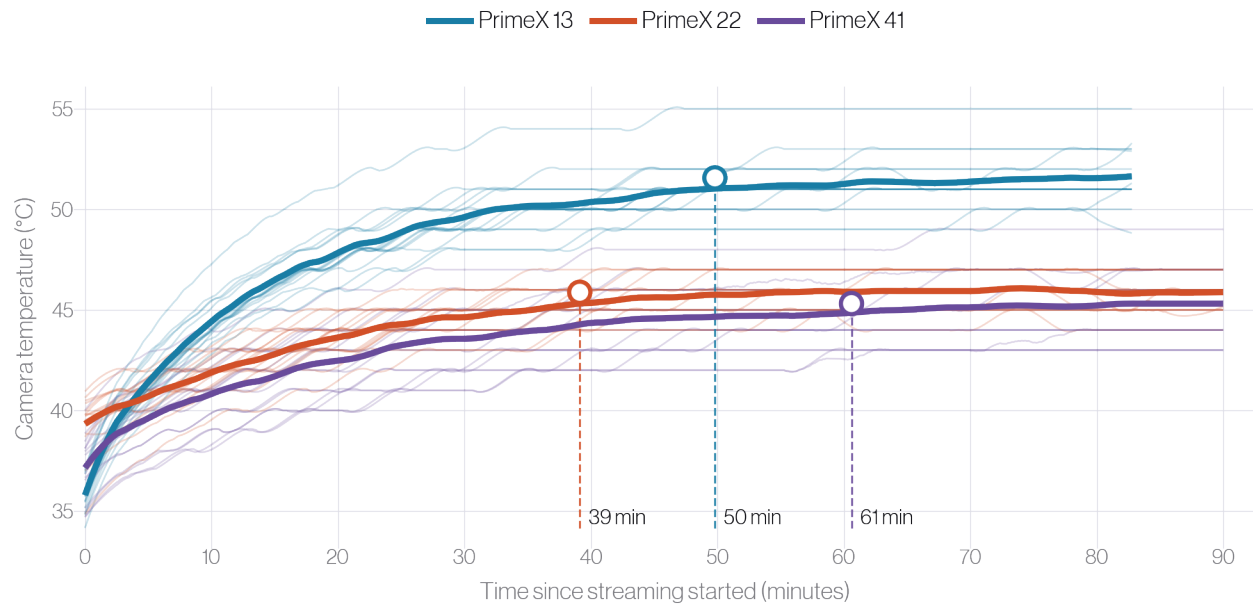


Figure 1: Warm-up for every camera model. Faint lines are individual cameras; the heavy line is the fleet average. The marked point is where that average reaches its stable temperature.

The PrimeX 13 settles hottest at 51.6 °C and the PrimeX 41 coolest at 45.3 °C, a spread of 6.2 °C between models. The PrimeX 13 climbs furthest, rising 14.8 °C from where it started.

Camera	Start	Stable	Rise	Time to stable	Time constant
PrimeX 13	36.8 °C	51.6 °C	14.8 °C	50 min	14 min
PrimeX 22	39.5 °C	45.9 °C	6.4 °C	39 min	19 min
PrimeX 41	37.6 °C	45.3 °C	7.8 °C	61 min	20 min

Cold Start

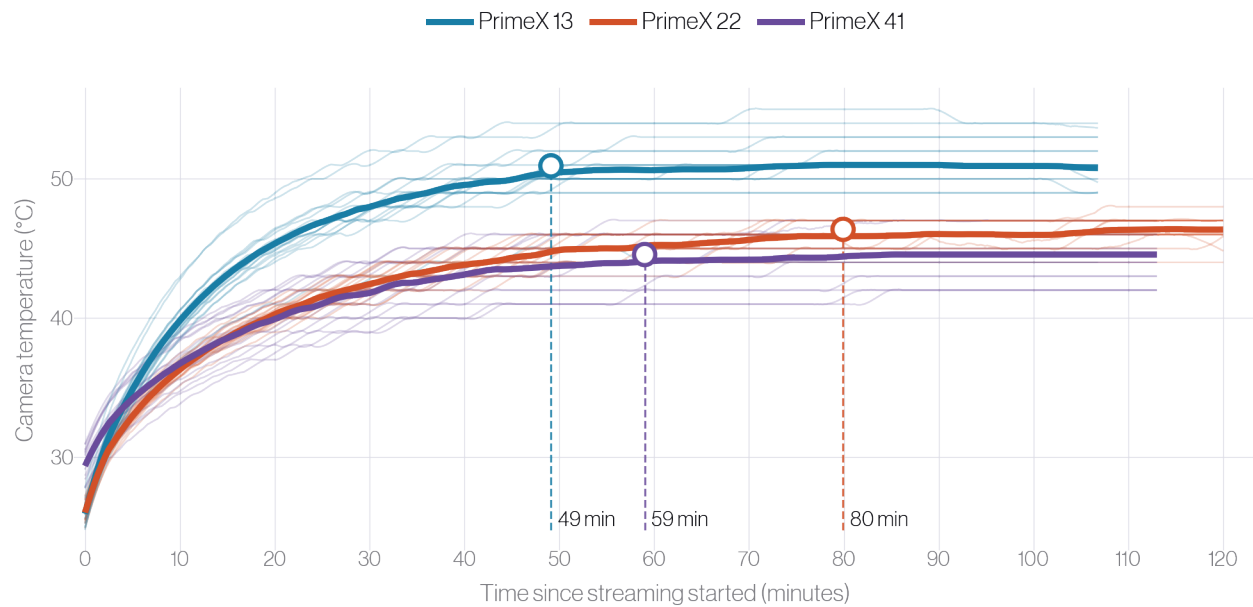


Figure 2: Warm-up for every camera model. Faint lines are individual cameras; the heavy line is the fleet average. The marked point is where that average reaches its stable temperature.

The PrimeX 13 settles hottest at 50.9 °C and the PrimeX 41 coolest at 44.6 °C, a spread of 6.4 °C between models. The PrimeX 13 climbs furthest, rising 23.6 °C from where it started.

Camera	Start	Stable	Rise	Time to stable	Time constant
PrimeX 13	27.3 °C	50.9 °C	23.6 °C	49 min	14 min
PrimeX 22	27.2 °C	46.4 °C	19.1 °C	80 min	18 min
PrimeX 41	30.2 °C	44.6 °C	14.4 °C	59 min	18 min

Spread between cameras

Run	Coollest unit	Hottest unit	Spread	Time constant
PrimeX 13	50 °C	55 °C	5 °C	13–16 min
PrimeX 13	44 °C	54 °C	10 °C	11–15 min
PrimeX 22	44 °C	47 °C	3 °C	17–23 min
PrimeX 22	44 °C	48 °C	4 °C	16–23 min
PrimeX 41	43 °C	49 °C	6 °C	15–23 min
PrimeX 41	42 °C	47 °C	5 °C	16–20 min

Individual cameras of the same model settle several degrees apart. This is normal and reflects position in the volume, airflow, and unit-to-unit variation.

By Model

Each faint line is one camera. The staircase shape is due to the camera's 1 °C temperature resolution. The group average, shown as a heavy line, smooths out these individual steps and provides a clearer view of the overall temperature trend.

PrimeX 13

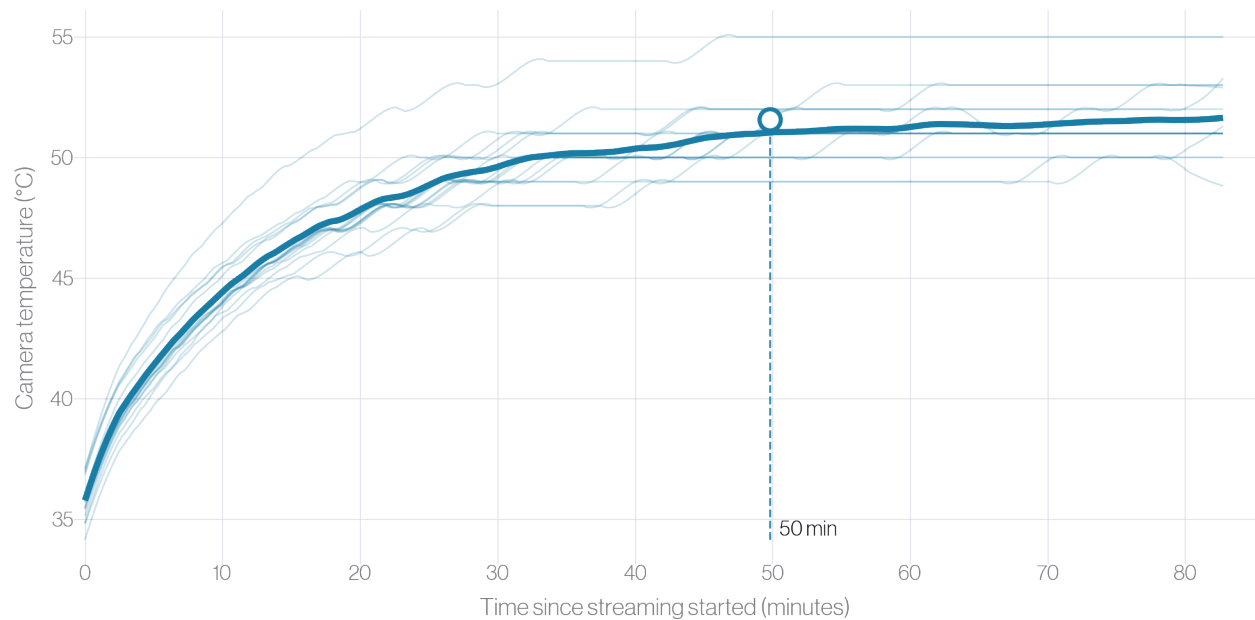


Figure 3: PrimeX 13, hibernation start. The camera group spans 5 °C. PrimeX 13 and PrimeX 13W were included in a single run, as only the lens differs.

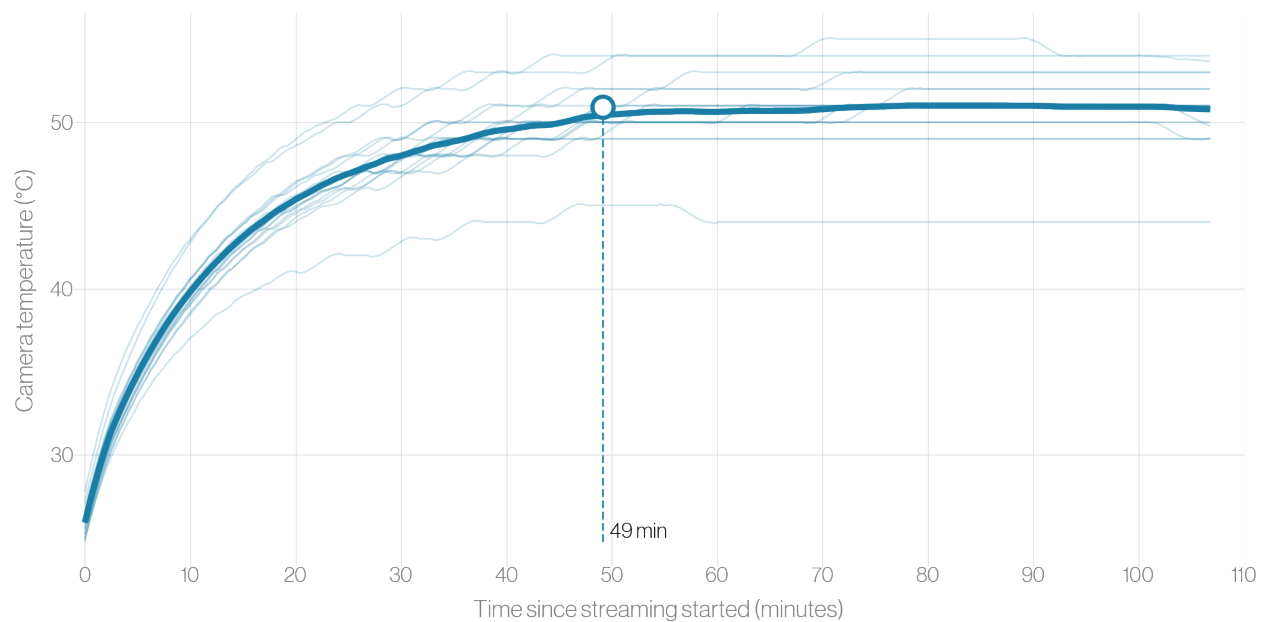


Figure 4: PrimeX 13, cold start. The camera group spans 10 °C. PrimeX 13 and PrimeX 13W were included in a single run, as only the lens differs.

PrimeX 22

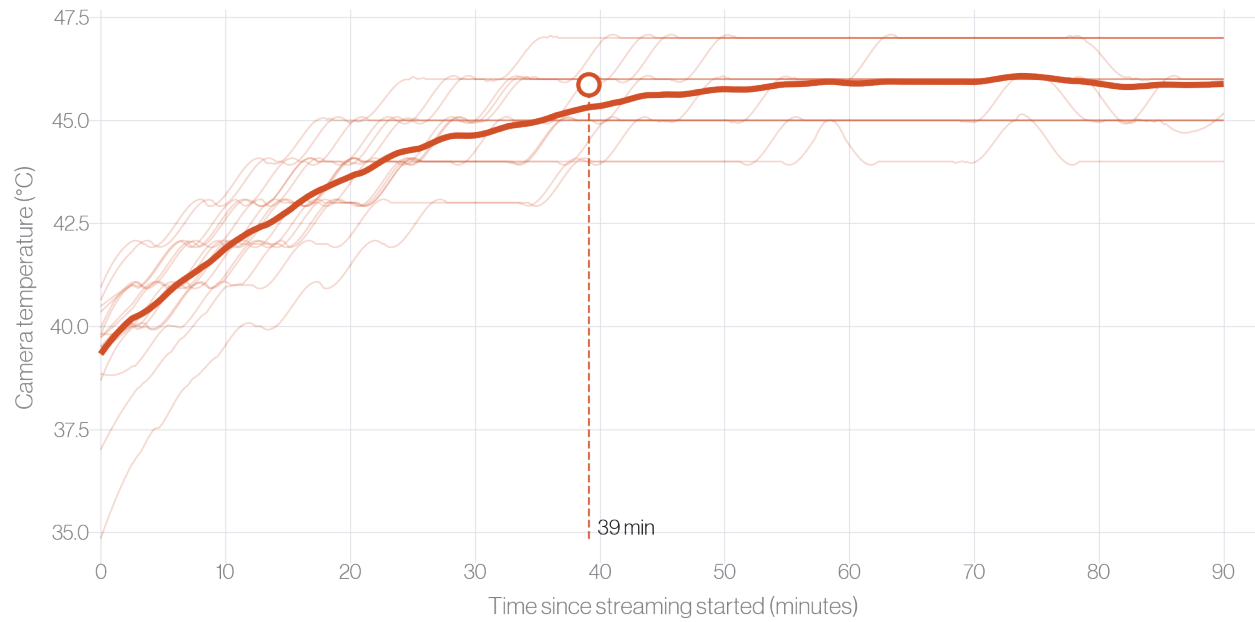


Figure 5: PrimeX 22, hibernation start. The camera group spans 3 °C.

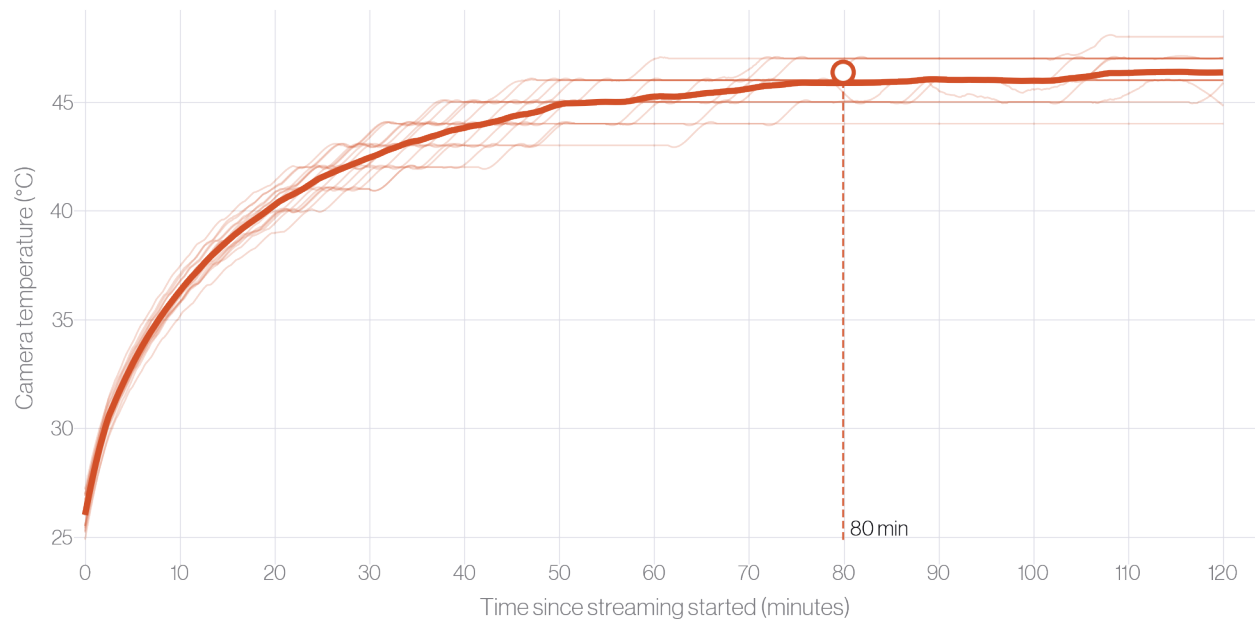


Figure 6: PrimeX 22, cold start. The camera group spans 4 °C.

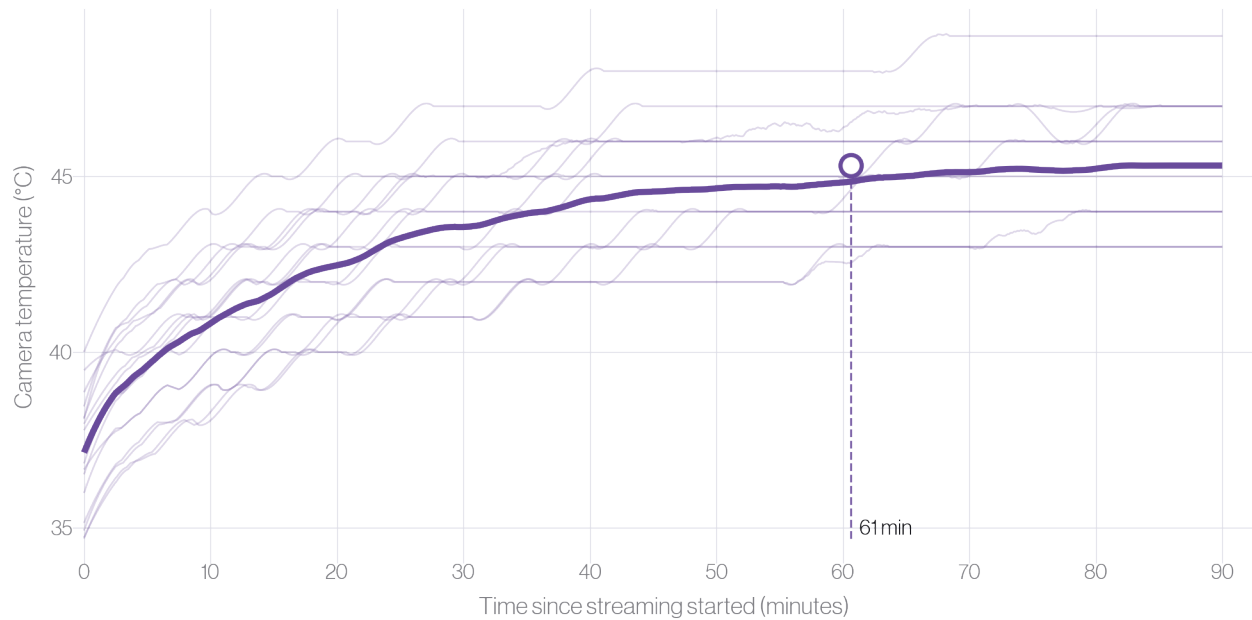
PrimeX 41

Figure 7: PrimeX 41, hibernation start. The camera group spans 6 °C.

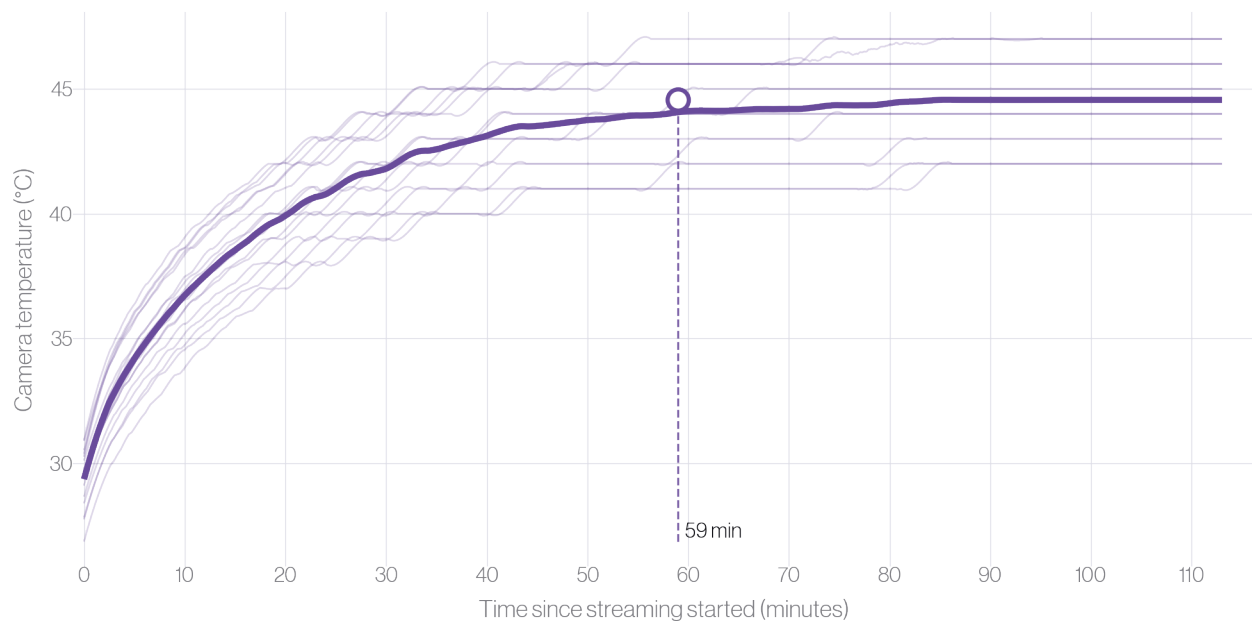


Figure 8: PrimeX 41, cold start. The camera group spans 5 °C.

How “stable” is defined

A camera is called stable at the earliest moment after which its temperature stays within 0.5 °C of its final settled level for the whole remainder of the run. For each model, this judgment is based on the aggregate of all cameras of that model.

Limitations

One ambient temperature, one frame rate. Everything here was recorded at 20 °C ambient in object mode at 120 Hz, with the ring light strobing in sync with exposure. Higher frame rates, exposure values and warmer rooms all raise the stable temperature, and may influence the time taken to reach it.

Passive Cooling. Cameras regulate heat through passive dissipation. Changes in environmental temperature during operation will affect the stability of the camera's own temperature.